

CSE 330 Assignment-3

Deadline: 04 January 2026

Question 1

Consider the function

$$f(x) = x^3 + x^2 - 4x - 4.$$

- (a) State the exact roots of $f(x)$. Construct two different fixed-point functions $g(x)$ such that solving $f(x) = 0$ is equivalent to finding a fixed point of $g(x)$.
- (b) Compute the convergence rate of each fixed-point function $g(x)$ obtained in part (a). Clearly state whether each iteration converges or diverges, and identify the root to which it converges (if applicable).

Question 2

Consider the function

$$f(x) = xe^x - 1.$$

- (a) Use Newton's method to find a solution of $f(x) = 0$ up to five iterations, starting from the initial guess $x_0 = 1.5$. Keep all results up to four significant figures.
- (b) Consider the fixed-point iteration

$$g(x) = \frac{2x + 1}{x + 1}.$$

Show that, for the method to be superlinearly convergent, the root must satisfy

$$x^* = -\frac{3}{2}.$$

Question 3

Consider the cubic function

$$f(x) = 2x^3 - 2x - 5.$$

Using Newton's method, compute a fixed-point function $g(x)$ that is superlinearly convergent for solving $f(x) = 0$.